

Kumar Thurimella, PhD

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EDUCATION & TRAINING

Stanford Health Care

Stanford, CA

Internal Medicine Residency, ABIM Research Pathway (Short Track)

Jun 2026–Jun 2028

- Guaranteed Rheumatology Fellowship, Stanford Health Care, Jul 2028–

University of Colorado, School of Medicine

Denver, CO

M.D. Candidate

Aug 2018–May 2026

- Clerkship Performance: Honors in 6/6 core clerkships (Internal Medicine, OB/GYN, Pediatrics, Family Medicine, Psychiatry, and Surgery), Rheumatology & Medicine Acting Internship; Elected to AOA

University of Cambridge

Cambridge, UK

Ph.D. in Biotechnology and Applied Statistics

Oct 2020–May 2024

- Advisors: Sergio Bacallado PhD (Cambridge), Ramnik J. Xavier MD/PhD (MGH/Broad Institute)
- Thesis: “Harnessing Deep Learning with Protein Language Models to Unveil Microbial Enzyme Function in Health and Disease”

University of Cambridge

Cambridge, UK

M.Phil. in Computational Biology (Biological Sciences), Advisor: Gosia Trynka PhD

Oct 2017–Aug 2018

University of Colorado, Boulder

Boulder, CO

B.S. in Applied Mathematics, *Magna Cum Laude*, Advisor: Rob Knight PhD

Aug 2009–Aug 2013

EXPERIENCE

Uber Technologies

San Francisco, CA

Software Engineer II

Mar 2015–Aug 2017

- Using a heuristic recursive backtracking algorithm, built an auto-scheduler to schedule all incoming candidates for Uber, that sits atop our own custom Applicant Tracking System
- Python, Flask, Tornado, React.js

ThoughtWorks

San Francisco, CA

Junior Software Developer

Apr 2014–Mar 2015

- Developed software as a consultant, building and maintaining web applications for clients
- Java, Spring

FIRST & CO-FIRST AUTHOR PUBLICATIONS

- **K. Thurimella[†], E. Wu[†]**, C. Li, D. B. Graham, R. M. Owens, D. R. Plichta, C. L. Sokol, R. J. Xavier, and S. Bacallado, “Identifying microbial protease allergens through protein language model-guided homology”, *Cell Systems*, vol. 17, no. 3, 2026.
- **K. Thurimella**, A. M. Mohamed, C. Li, T. Vatanen, D. B. Graham, R. M. Owens, S. L. La Rosa, D. R. Plichta, S. Bacallado, and R. J. Xavier, “Protein language models uncover carbohydrate-active enzyme function in metagenomics”, *BMC Bioinformatics*, vol. 26, no. 285, Nov. 2025.
- **A. Xia[†], K. Thurimella[†]**, S. Bull, J. Waughtal, C. Chavez, S. Novins-Montague, J. Silvasstar, A. Salyers, M. P. Ho, and M. Laveri, “Pilot randomized controlled trial: Increased engagement in cardiovascular health through ai-enabled tailored messaging”, *JMIR Cardio*, Jan. 2024, *Under Review*.
- **M. Shaffer[†], K. Thurimella[†]**, J. D. Sterrett, and C. A. Lozupone, “Scnic: Sparse correlation network investigation for compositional data”, *Molecular Ecology Resources*, vol. 23, no. 1, pp. 312–325, Jan. 2023.

CONTRIBUTING AUTHOR PUBLICATIONS

- V. Stoeger, M. Strauss, **K. Thurimella**, ..., C. Gerner, and R. M. Owens, “Bioelectronic technology for nutritional research—a novel in vitro platform for a better understanding of human gut barrier absorption”, *Advanced Biology*, vol. 10, no. 2, e00409, Feb. 2026.
- Y.-M. Hung, Y.-S. Hyun, R. Cohen, M.-M. Pust, **K. Thurimella**, G. Pishchany, H. Lau, M. Stražar, R. J. Xavier, and D. B. Graham, “Commensal bacterial toxins elicit dynamic cd4 t cell responses that shift between tolerance and inflammation”, *Science Immunology*, 2025, *Under Review*.
- K. S. Jabbar, S. Priya, J. Xu, U. Das Adhikari, G. Pishchany, J. Johansen, **K. Thurimella**, ..., D. R. Plichta, D. S. Kwon, and R. J. Xavier, “Antiretroviral drugs modulate systemic inflammation through direct interactions with gut microbial strains”, *Nature Microbiology*, Oct. 2025.
- M. Kadoki, K. Nanki, D. Szamosvári, M. Stražar, **K. Thurimella**, ..., D. B. Graham, and R. J. Xavier, “Decoding host–environment chemosensory dialogues through intestinal G protein-coupled receptors”, *Cell Host & Microbe*, 2025, *Under Review*.
- C. Li, M. Stražar, A. M. T. Mohamed, ..., **K. Thurimella**, ..., D. R. Plichta, and R. J. Xavier, “Gut microbiome and metabolome profiling in framingham heart study reveals cholesterol-metabolizing bacteria linked with lower cardiovascular risk”, *Cell*, Mar. 2024.
- L. Wells, **K. Thurimella**, and S. Bacallado, “Regularised canonical correlation analysis: Graphical lasso, biplots and beyond”, *Journal of the American Statistical Association*, Oct. 2023, *Under Review*.
- M. Shaffer, **K. Thurimella**, K. Quinn, K. Doenges, X. Zhang, S. Bokatzian, N. Reisdorph, and C. A. Lozupone, “Amon: Annotation of metabolite origins via networks to integrate microbiome and metabolome data”, *BMC bioinformatics*, vol. 20, no. 1, pp. 1–11, Nov. 2019.

† denotes co-first authorship.

PRESENTATIONS

- K. Thurimella, E.C. Wu, C. Li, D.R. Plichta, C.L. Sokol, R.J. Xavier, S. Bacallado, Protein Language Model Guided Discovery of Microbial Protease Allergens (Oral), Gordon Research Conference 2026, Ventura
- K. Thurimella, C. Li, A.M. Mohamed, D.B. Graham, R. Owens, S.L.L Rosa, C.L. Sokol, D.R. Plichta, S. Bacallado, R.J. Xavier, Identifying Putative Microbial Enzyme Candidates Linked with Allergy and Glycobiology through Protein Language Model-Guided Homology, Keystone Symposia Microbiota and Cancer Immunity 2024, Taipei
- K. Thurimella, M. Shaffer, C. Lozupone, Employing Metabolomics and Microbiome Data to Build Algorithm for Interrogating Host-Microbe Interactions (Oral), Western Medical Research Conference 2020, Carmel

SCHOLARSHIPS AND AWARDS

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| • Gordon Research Conference, Carl Storm Fellow (Full Registration & Accommodation) | 2025 |
| • Student Clinical Professionalism Award | 2025 |
| • Keystone Symposia Future of Science Scholarship | 2024 |
| • Gates Cambridge Scholarship | 2020–2024 |
| – Combining Maths and Medicine | |
| – CU Medical Student Wins \$250,000 Scholarship | |
| • Rotary International Scholarship | 2020–2024 |
| • Dr. Henry Christian Award, American Federation for Medical Research - Most Outstanding Abstract | 2019 |
| • Wellcome Trust Departmental Award Full Scholarship, University of Cambridge | 2017 |
| • Norlin Scholars Scholarship | 2011–2013 |
| • Engineering Merit Scholarship | 2009–2013 |